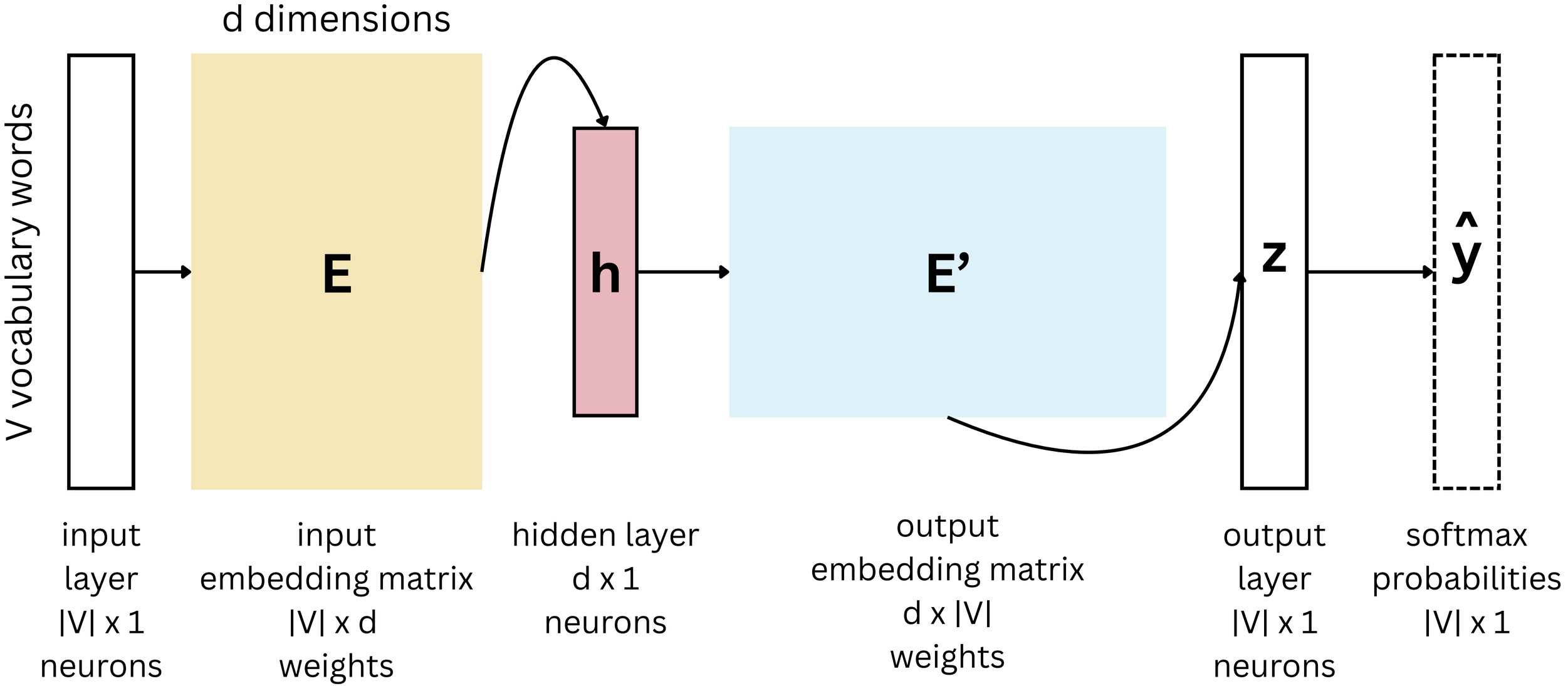
Natural Language Processing
Lab 4

Lonny Chen: l.chen@hertie-school.org
GRAD-E1282, Fall 2026
23-24 September 2026

Today’s Agenda

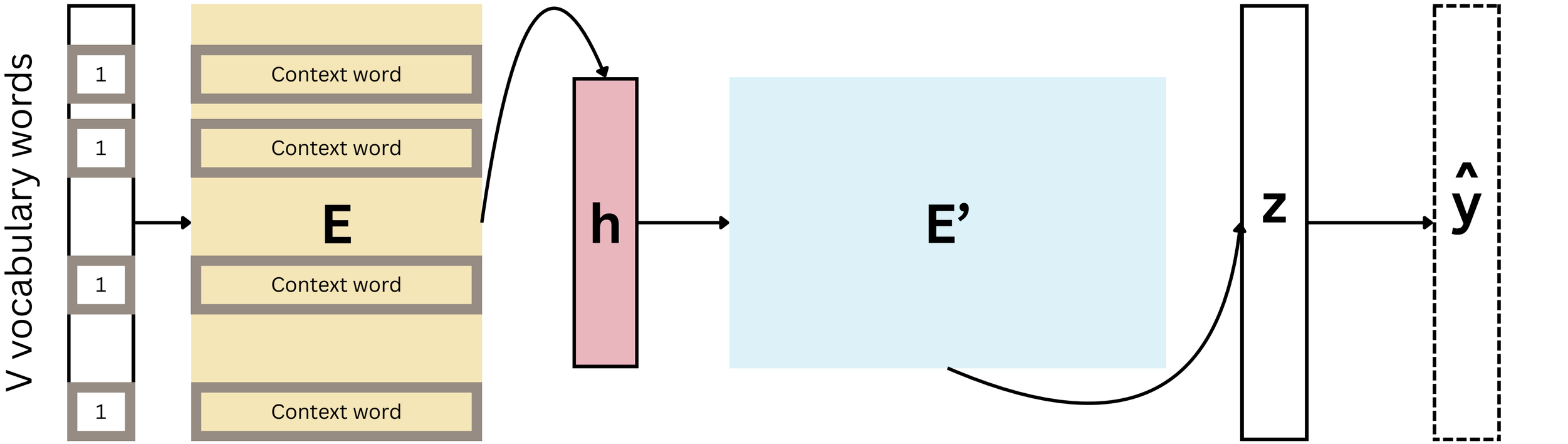
1. Continuous Bag-of-Words (CBoW)
2. Pre-Trained Embedding Model
3. Document Embeddings and Application
4. Custom Embedding Model

CBoW - Neural Network Architecture



CBoW - Neural Network Architecture

1 LOOK UP d dimensions



input layer |V| x 1 neurons

input embedding matrix |V| x d weights

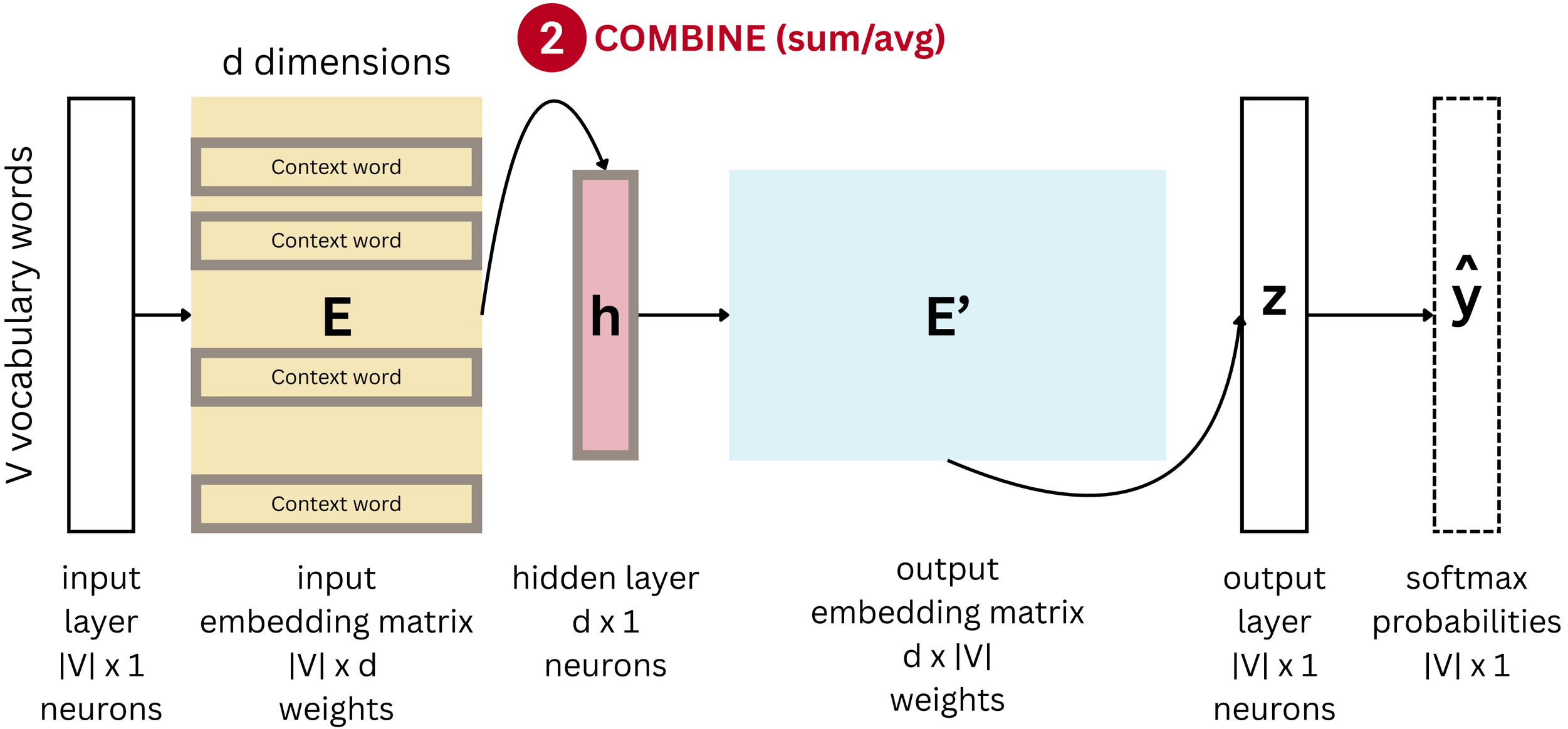
hidden layer d x 1 neurons

output embedding matrix d x |V| weights

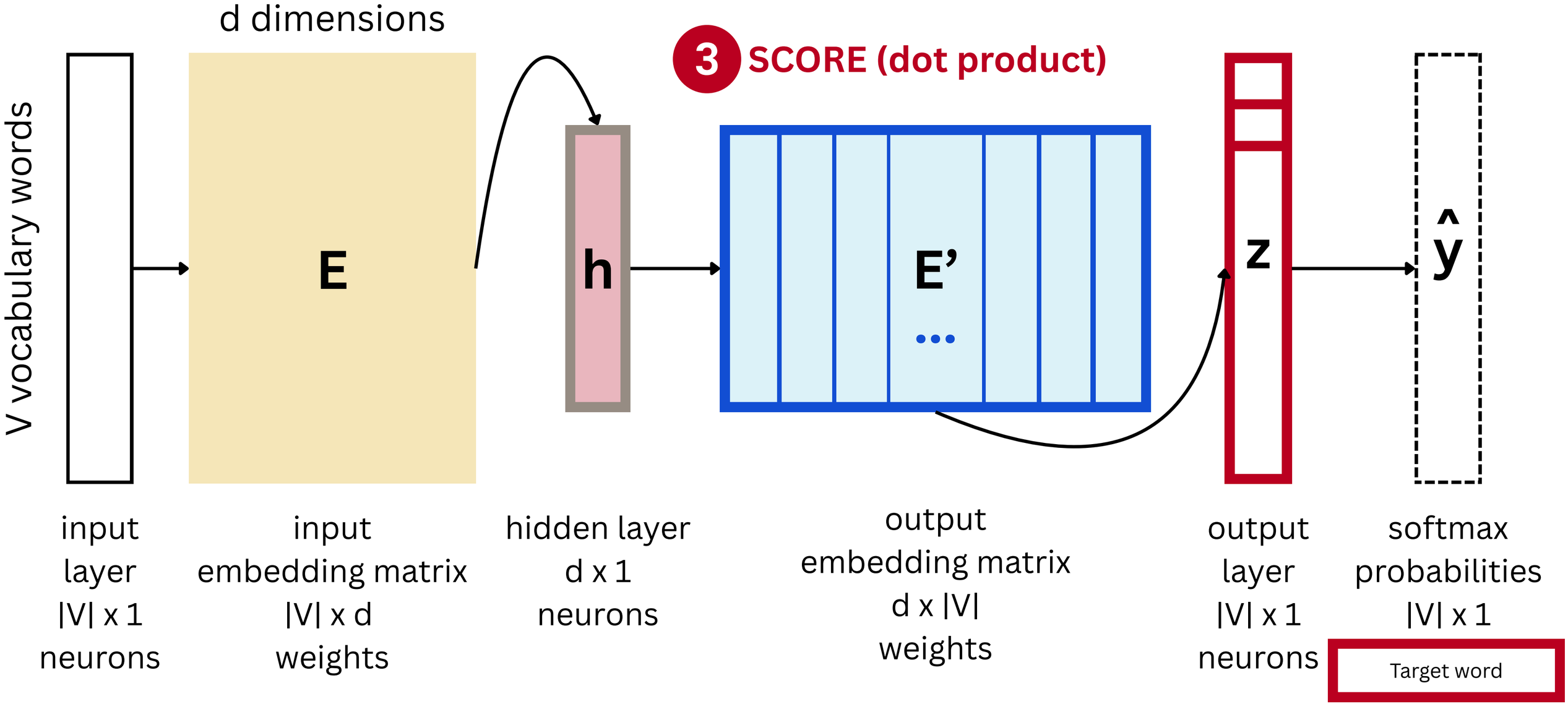
output layer |V| x 1 neurons

softmax probabilities |V| x 1

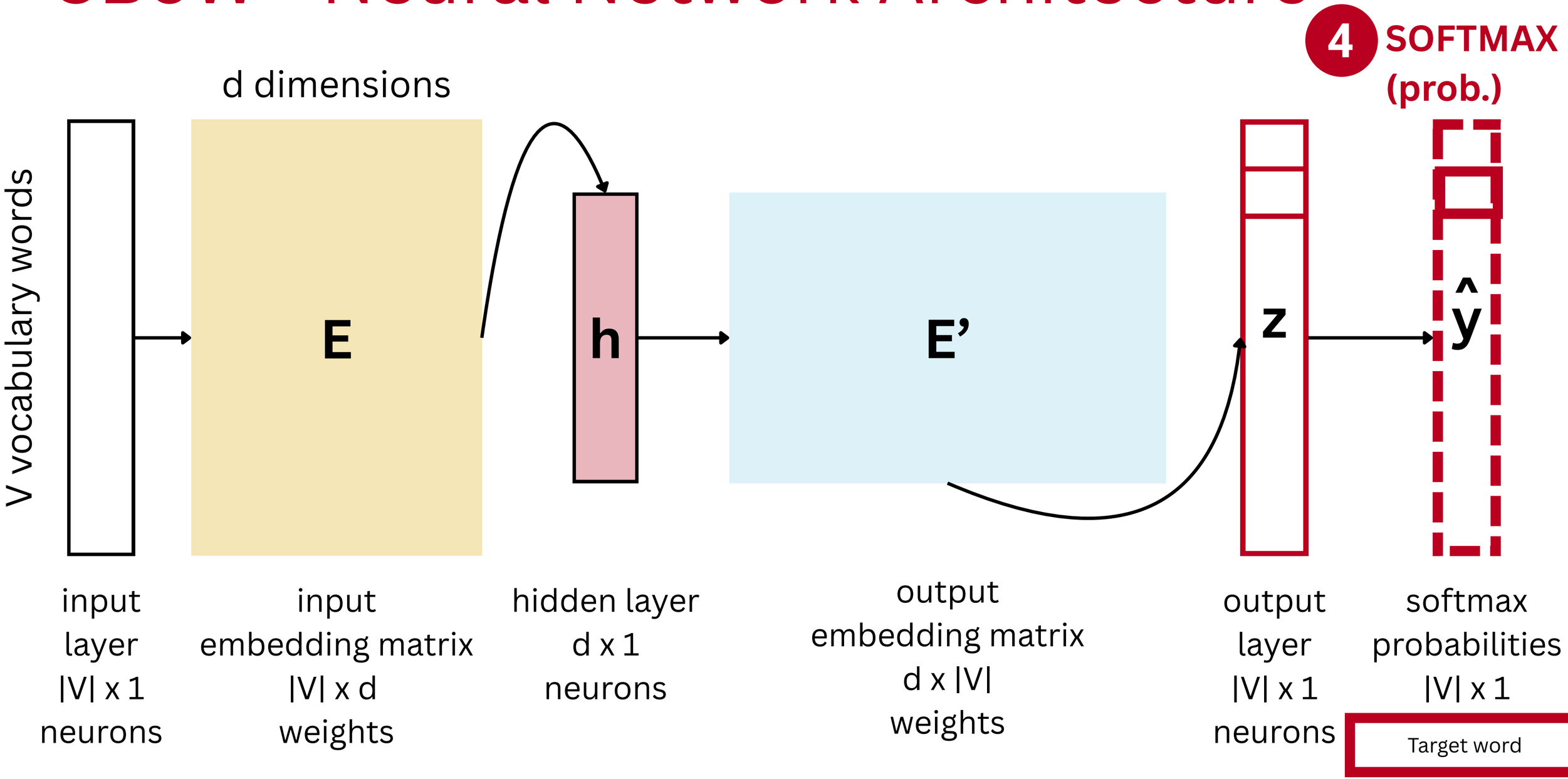
CBoW - Neural Network Architecture



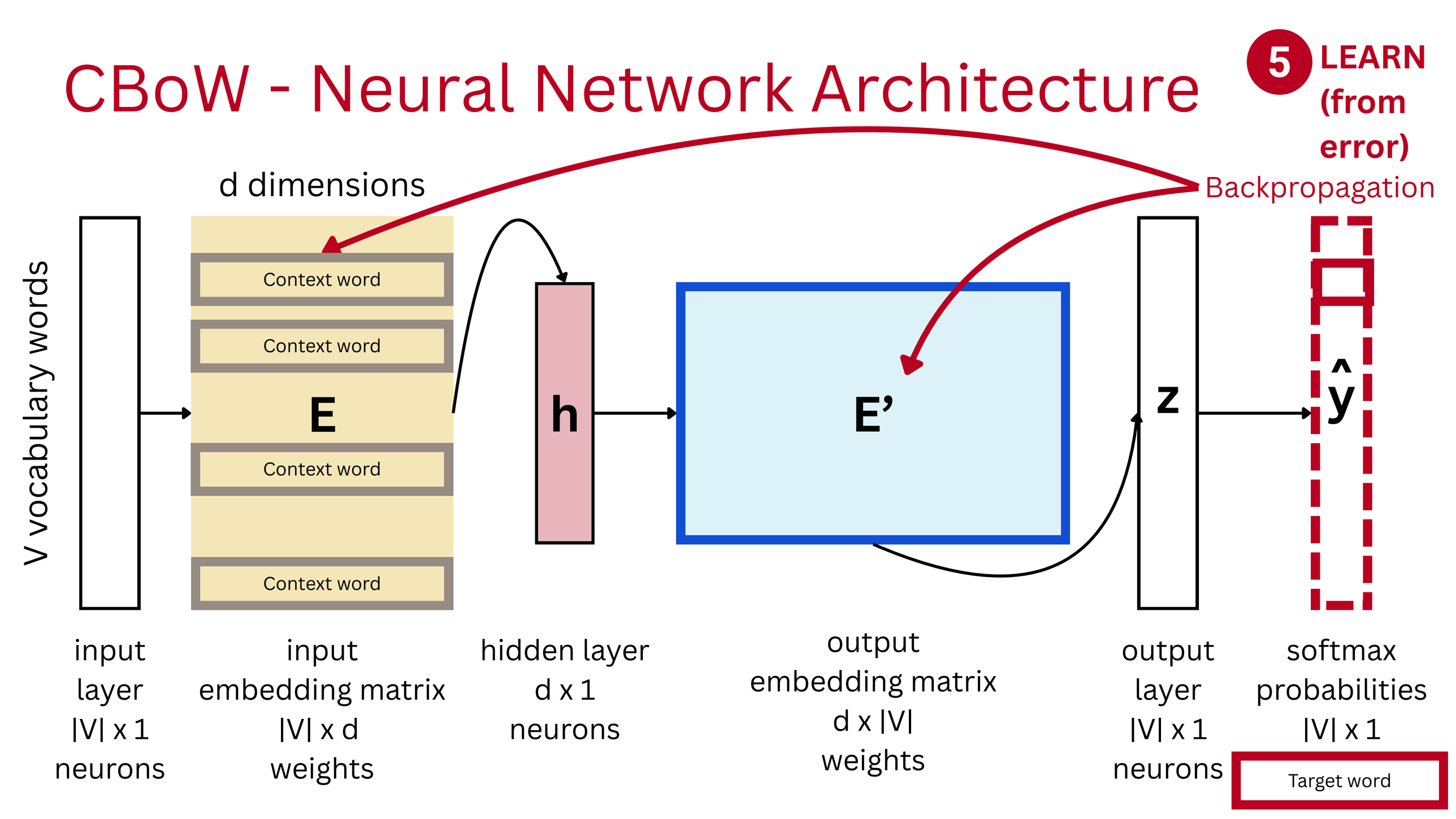
CBoW - Neural Network Architecture



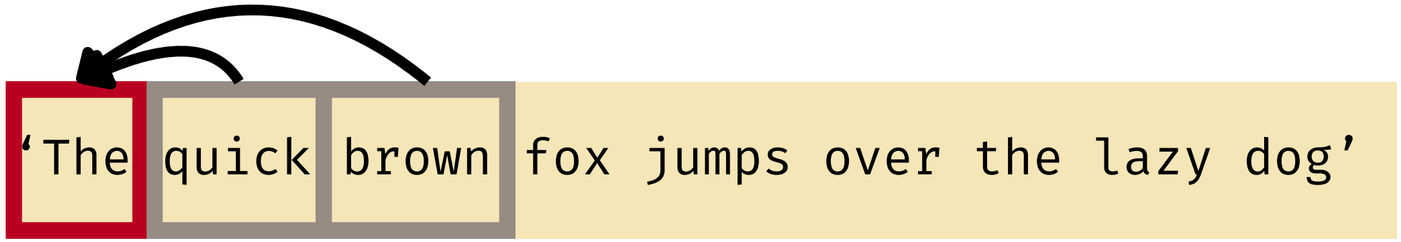
CBoW - Neural Network Architecture



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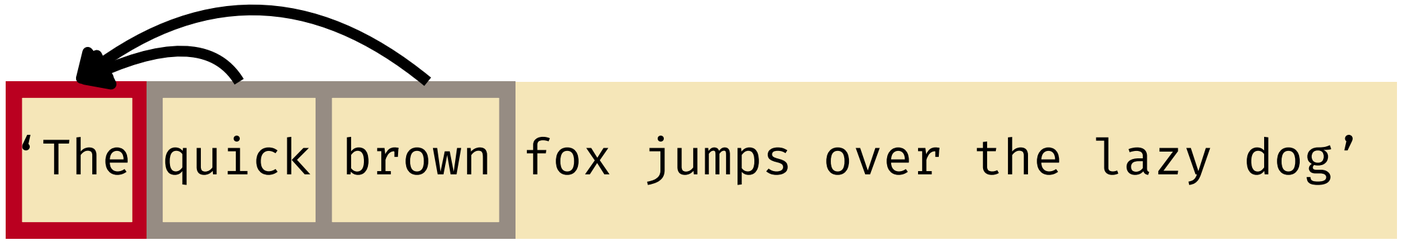
CBoW - Forward Pass



	E input													
	E_0	E_1	E_2	E_3	E_4	E'_0	E'_1	E'_2	E'_3	E'_4	scores	y_hat	y_actual	error
**the	-0.13	0.45	0.23	0.10	-0.34	-0.38	-0.00	-0.47	0.41	-0.24	-0.03	0.13	1.0	-0.87
*quick	-0.34	-0.44	0.37	0.10	0.21	0.16	-0.19	0.02	0.05	-0.32	-0.05	0.13	0.0	0.13
*brown	-0.48	0.47	0.33	-0.29	-0.32	0.47	0.28	0.44	0.39	0.10	-0.08	0.12	0.0	0.12
fox	-0.32	-0.20	0.02	-0.07	-0.21	0.42	-0.41	-0.30	-0.45	-0.17	-0.23	0.11	0.0	0.11
jumps	0.11	-0.36	-0.21	-0.13	-0.04	-0.11	-0.23	0.33	-0.14	-0.22	0.18	0.16	0.0	0.16
over	0.29	-0.30	0.01	0.09	-0.45	0.04	-0.36	0.30	-0.43	0.49	0.10	0.15	0.0	0.15
lazy	0.11	-0.33	-0.43	0.45	0.47	0.27	-0.30	-0.49	0.32	0.21	-0.33	0.10	0.0	0.10
dog	0.31	-0.20	-0.40	0.18	-0.06	0.23	0.27	-0.43	-0.14	-0.38	-0.21	0.11	0.0	0.11

1 LOOK UP C context words in E input embedding (row vectors)

CBoW - Forward Pass



E input

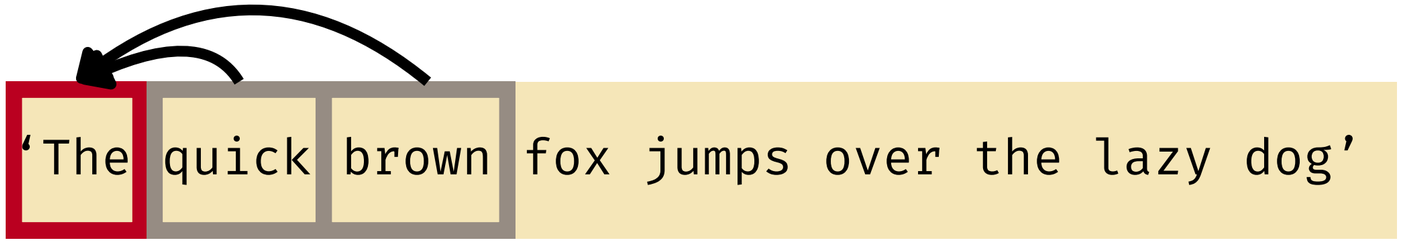
[-0.41 0.01 0.35 -0.09 -0.06]

$$\mathbf{h} = \frac{1}{C}(\mathbf{E}_1 + \cdots + \mathbf{E}_C)$$

	E_0	E_1	E_2	E_3	E_4	E'_0	E'_1	E'_2	E'_3	E'_4	scores	y_hat	y_actual	error
**the	-0.13	0.45	0.23	0.10	-0.34	-0.38	-0.00	-0.47	0.41	-0.24	-0.03	0.13	1.0	-0.87
*quick	-0.34	-0.44	0.37	0.10	0.21	0.16	-0.19	0.02	0.05	-0.32	-0.05	0.13	0.0	0.13
*brown	-0.48	0.47	0.33	-0.29	-0.32	0.47	0.28	0.44	0.39	0.10	-0.08	0.12	0.0	0.12
fox	-0.32	-0.20	0.02	-0.07	-0.21	0.42	-0.41	-0.30	-0.45	-0.17	-0.23	0.11	0.0	0.11
jumps	0.11	-0.36	-0.21	-0.13	-0.04	-0.11	-0.23	0.33	-0.14	-0.22	0.18	0.16	0.0	0.16
over	0.29	-0.30	0.01	0.09	-0.45	0.04	-0.36	0.30	-0.43	0.49	0.10	0.15	0.0	0.15
lazy	0.11	-0.33	-0.43	0.45	0.47	0.27	-0.30	-0.49	0.32	0.21	-0.33	0.10	0.0	0.10
dog	0.31	-0.20	-0.40	0.18	-0.06	0.23	0.27	-0.43	-0.14	-0.38	-0.21	0.11	0.0	0.11

2 COMBINE (sum or average) context vectors as “context summary” at hidden layer

CBoW - Forward Pass



[−0.41 0.01 0.35 −0.09 −0.06]

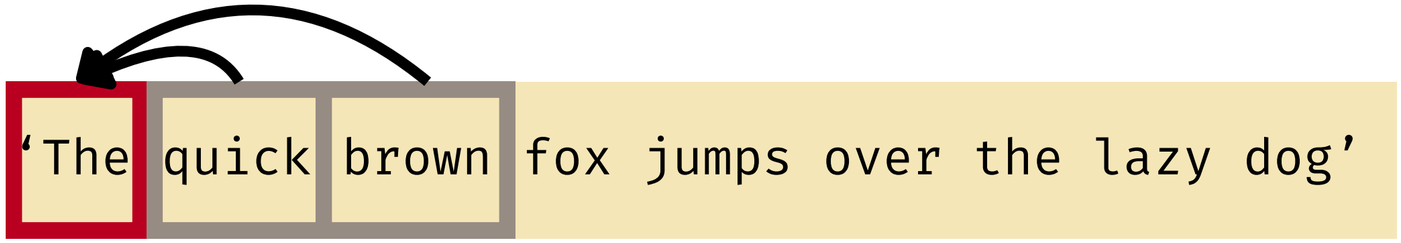
$$z_j = \mathbf{h} \cdot \mathbf{E'}_j$$

	E_0	E_1	E_2	E_3	E_4	E'_0	E'_1	E'_2	E'_3	E'_4	scores	y_hat	y_actual	error
**the	-0.13	0.45	0.23	0.10	-0.34	-0.38	-0.00	-0.47	0.41	-0.24	-0.03	0.13	1.0	-0.87
*quick	-0.34	-0.44	0.37	0.10	0.21	0.16	-0.19	0.02	0.05	-0.32	-0.05	0.13	0.0	0.13
*brown	-0.48	0.47	0.33	-0.29	-0.32	0.47	0.28	0.44	0.39	0.10	-0.08	0.12	0.0	0.12
fox	-0.32	-0.20	0.02	-0.07	-0.21	0.42	-0.41	-0.30	-0.45	-0.17	-0.23	0.11	0.0	0.11
jumps	0.11	-0.36	-0.21	-0.13	-0.04	-0.11	-0.23	0.33	-0.14	-0.22	0.18	0.16	0.0	0.16
over	0.29	-0.30	0.01	0.09	-0.45	0.04	-0.36	0.30	-0.43	0.49	0.10	0.15	0.0	0.15
lazy	0.11	-0.33	-0.43	0.45	0.47	0.27	-0.30	-0.49	0.32	0.21	-0.33	0.10	0.0	0.10
dog	0.31	-0.20	-0.40	0.18	-0.06	0.23	0.27	-0.43	-0.14	-0.38	-0.21	0.11	0.0	0.11

E’ output

3 SCORE “fit” (dot product) of context to each output word j’s vector in E’

CBoW - Forward Pass

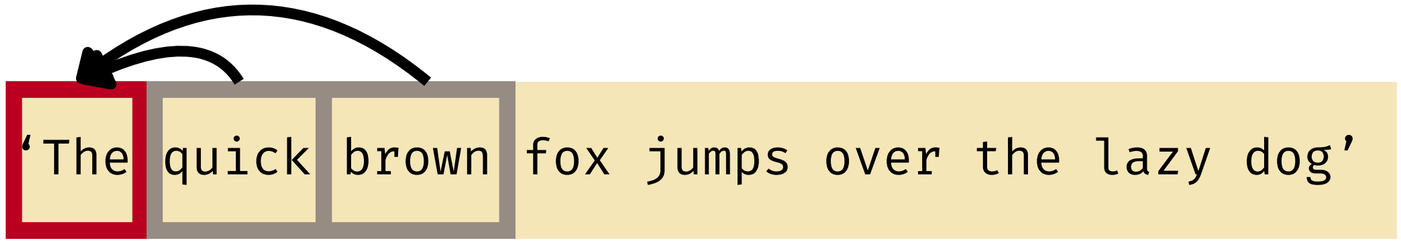


$$\hat{y}_j = \text{softmax}(z_j) = \frac{e^{z_j}}{\sum_{j=1}^{V} e^{z_j}}$$

	E_0	E_1	E_2	E_3	E_4	E'_0	E'_1	E'_2	E'_3	E'_4	scores	y_hat	y_actual	error
**the	-0.13	0.45	0.23	0.10	-0.34	-0.38	-0.00	-0.47	0.41	-0.24	-0.03	0.13	1.0	-0.87
*quick	-0.34	-0.44	0.37	0.10	0.21	0.16	-0.19	0.02	0.05	-0.32	-0.05	0.13	0.0	0.13
*brown	-0.48	0.47	0.33	-0.29	-0.32	0.47	0.28	0.44	0.39	0.10	-0.08	0.12	0.0	0.12
fox	-0.32	-0.20	0.02	-0.07	-0.21	0.42	-0.41	-0.30	-0.45	-0.17	-0.23	0.11	0.0	0.11
jumps	0.11	-0.36	-0.21	-0.13	-0.04	-0.11	-0.23	0.33	-0.14	-0.22	0.18	0.16	0.0	0.16
over	0.29	-0.30	0.01	0.09	-0.45	0.04	-0.36	0.30	-0.43	0.49	0.10	0.15	0.0	0.15
lazy	0.11	-0.33	-0.43	0.45	0.47	0.27	-0.30	-0.49	0.32	0.21	-0.33	0.10	0.0	0.10
dog	0.31	-0.20	-0.40	0.18	-0.06	0.23	0.27	-0.43	-0.14	-0.38	-0.21	0.11	0.0	0.11

4 SOFTMAX output scores into a probability (∑=1) for each output word j

CBoW - Forward Pass

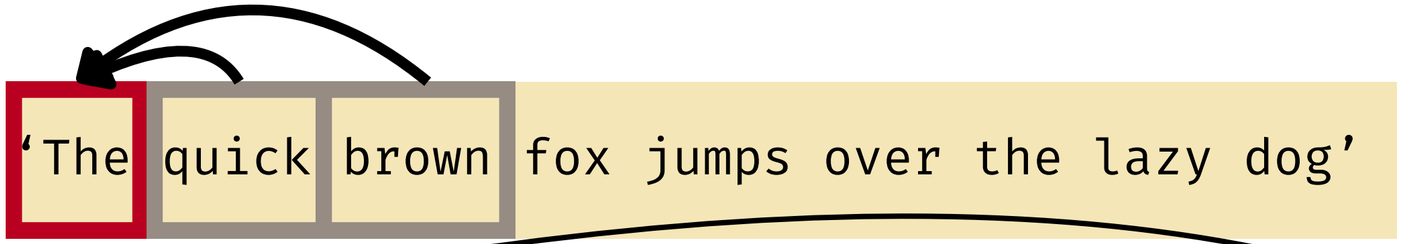


	E_0	E_1	E_2	E_3	E_4	E'_0	E'_1	E'_2	E'_3	E'_4	scores	y_hat	y_actual	error
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*quick	-0.34	-0.44	0.37	0.10	0.21	0.16	-0.19	0.02	0.05	-0.32	-0.05	0.13	0.0	0.13
*brown	-0.48	0.47	0.33	-0.29	-0.32	0.47	0.28	0.44	0.39	0.10	-0.08	0.12	0.0	0.12
fox	-0.32	-0.20	0.02	-0.07	-0.21	0.42	-0.41	-0.30	-0.45	-0.17	-0.23	0.11	0.0	0.11
jumps	0.11	-0.36	-0.21	-0.13	-0.04	-0.11	-0.23	0.33	-0.14	-0.22	0.18	0.16	0.0	0.16
over	0.29	-0.30	0.01	0.09	-0.45	0.04	-0.36	0.30	-0.43	0.49	0.10	0.15	0.0	0.15
lazy	0.11	-0.33	-0.43	0.45	0.47	0.27	-0.30	-0.49	0.32	0.21	-0.33	0.10	0.0	0.10
dog	0.31	-0.20	-0.40	0.18	-0.06	0.23	0.27	-0.43	-0.14	-0.38	-0.21	0.11	0.0	0.11

$$\frac{\partial L}{\partial z_j} = e_j = \hat{y}_j - y_j$$

5 **LEARN from prediction error (partial derivative of loss function over output)**

CBoW - Backward Pass



[-0.41 0.01 0.35 -0.09 -0.06]

E’ output

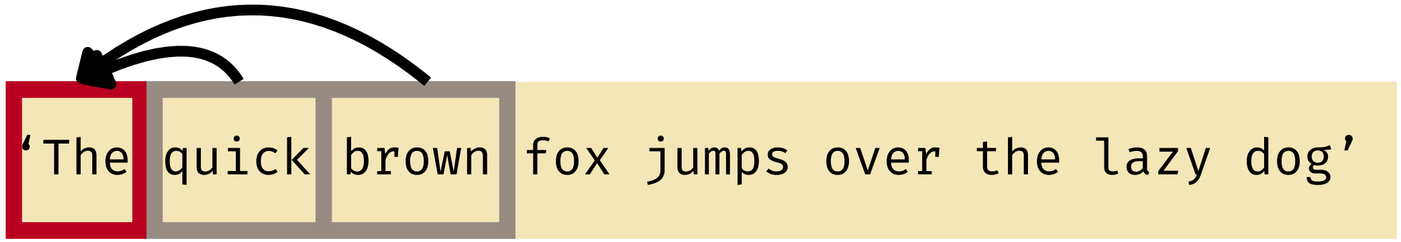
	E_0	E_1	E_2	E_3	E_4	E'_0	E'_1	E'_2	E'_3	E'_4	scores	y_hat	y_actual	error
**the	-0.13	0.45	0.23	0.10	-0.34	-0.38	-0.00	-0.47	0.41	-0.24	-0.03	0.13	1.0	-0.87
*quick	-0.34	-0.44	0.37	0.10	0.21	0.16	-0.19	0.02	0.05	-0.32	-0.05	0.13	0.0	0.13
*brown	-0.48	0.47	0.33	-0.29	-0.32	0.47	0.28	0.44	0.39	0.10	-0.08	0.12	0.0	0.12
fox	-0.32	-0.20	0.02	-0.07	-0.21	0.42	-0.41	-0.30	-0.45	-0.17	-0.23	0.11	0.0	0.11
jumps	0.11	-0.36	-0.21	-0.13	-0.04	-0.11	-0.23	0.33	-0.14	-0.22	0.18	0.16	0.0	0.16
over	0.29	-0.30	0.01	0.09	-0.45	0.04	-0.36	0.30	-0.43	0.49	0.10	0.15	0.0	0.15
lazy	0.11	-0.33	-0.43	0.45	0.47	0.27	-0.30	-0.49	0.32	0.21	-0.33	0.10	0.0	0.10
dog	0.31	-0.20	-0.40	0.18	-0.06	0.23	0.27	-0.43	-0.14	-0.38	-0.21	0.11	0.0	0.11

$$\frac{\partial L}{\partial \mathbf{E'}_j} = e_j \mathbf{h}$$

$$\mathbf{E'}_j^{\text{new}} = \mathbf{E'}_j^{\text{old}} - \eta(e_j \mathbf{h})$$

1 **UPDATE each output vector in E’ with a portion of h weighted by its error**

CBoW - Backward Pass



[-0.41 0.01 0.35 -0.09 -0.06]

E’ output

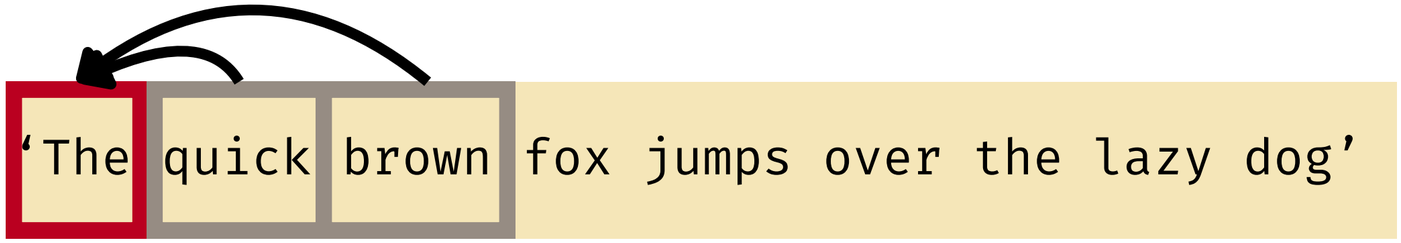
	E_0	E_1	E_2	E_3	E_4	E'_0	E'_1	E'_2	E'_3	E'_4	scores	y_hat	y_actual	error
**the	-0.13	0.45	0.23	0.10	-0.34	-0.56	0.01	-0.32	0.37	-0.26	-0.03	0.13	1.0	-0.87
*quick	-0.46	-0.41	0.26	0.21	0.17	0.19	-0.19	-0.00	0.06	-0.32	-0.05	0.13	0.0	0.13
*brown	-0.60	0.50	0.22	-0.18	-0.36	0.50	0.28	0.42	0.40	0.10	-0.08	0.12	0.0	0.12
fox	-0.32	-0.20	0.02	-0.07	-0.21	0.44	-0.41	-0.32	-0.44	-0.17	-0.23	0.11	0.0	0.11
jumps	0.11	-0.36	-0.21	-0.13	-0.04	-0.08	-0.23	0.30	-0.13	-0.22	0.18	0.16	0.0	0.16
over	0.29	-0.30	0.01	0.09	-0.45	0.07	-0.36	0.27	-0.42	0.49	0.10	0.15	0.0	0.15
lazy	0.11	-0.33	-0.43	0.45	0.47	0.29	-0.30	-0.51	0.32	0.21	-0.33	0.10	0.0	0.10
dog	0.31	-0.20	-0.40	0.18	-0.06	0.25	0.27	-0.45	-0.13	-0.38	-0.21	0.11	0.0	0.11

$$\frac{\partial L}{\partial \mathbf{E'}_j} = e_j \mathbf{h}$$

$$\mathbf{E'}_j^{\text{new}} = \mathbf{E'}_j^{\text{old}} - \eta(e_j \mathbf{h})$$

1 **UPDATE each output vector in E’ with a portion of h weighted by its error**

CBoW - Backward Pass



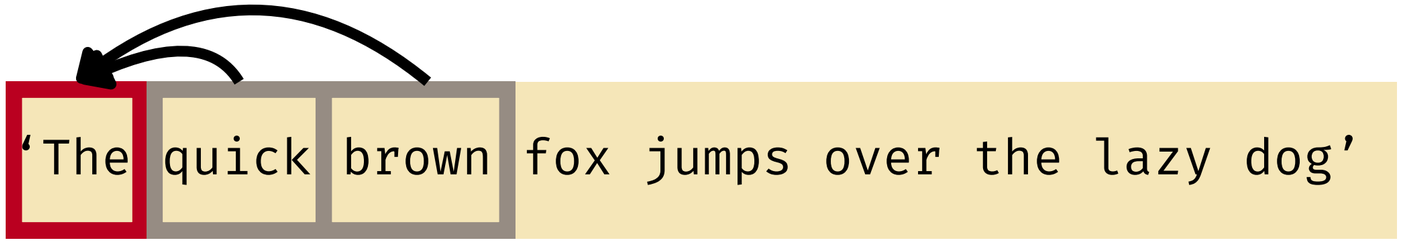
	E input					E’ output								
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$$\frac{\partial L}{\partial \mathbf{E}_k} = \frac{1}{C} \sum_{j=1}^{V} e_j \mathbf{E'}_j$$

$$\mathbf{E}_k^{\text{new}} = \mathbf{E}_k^{\text{old}} - \eta \left(\frac{1}{C} \sum_{j=1}^{V} e_j \mathbf{E}'_j \right)$$

2 UPDATE context input k vectors in E with the SUM of each E’ output @ its error

CBoW - Backward Pass



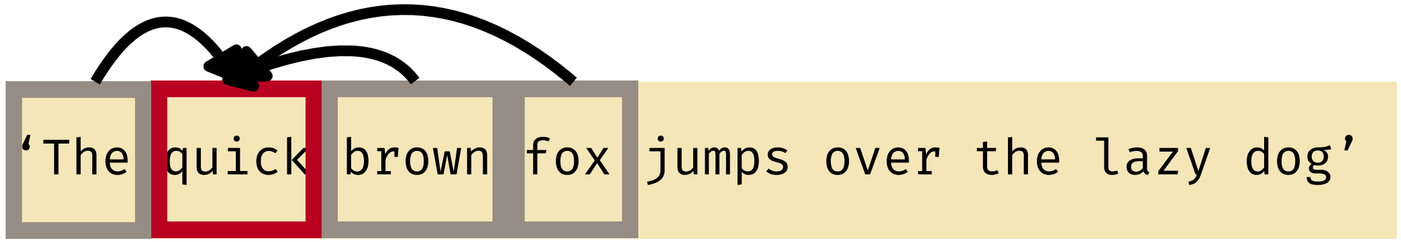
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	E_0	E_1	E_2	E_3	E_4	E'_0	E'_1	E'_2	E'_3	E'_4	scores	y_hat	y_actual	error
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jumps	0.11	-0.36	-0.21	-0.13	-0.04	-0.08	-0.23	0.30	-0.13	-0.22	0.18	0.16	0.0	0.16
over	0.29	-0.30	0.01	0.09	-0.45	0.07	-0.36	0.27	-0.42	0.49	0.10	0.15	0.0	0.15
lazy	0.11	-0.33	-0.43	0.45	0.47	0.29	-0.30	-0.51	0.32	0.21	-0.33	0.10	0.0	0.10
dog	0.31	-0.20	-0.40	0.18	-0.06	0.25	0.27	-0.45	-0.13	-0.38	-0.21	0.11	0.0	0.11

$$\frac{\partial L}{\partial \mathbf{E}_k} = \frac{1}{C} \sum_{j=1}^{V} e_j \mathbf{E'}_j$$

$$\mathbf{E}_k^{\text{new}} = \mathbf{E}_k^{\text{old}} - \eta(\frac{1}{C} \sum_{j=1}^{V} e_j \mathbf{E}'_j)$$

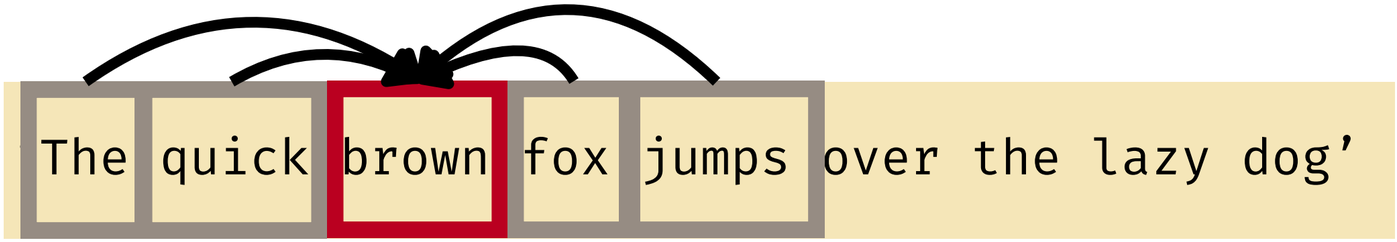
2 UPDATE context input k vectors in E with the SUM of each E’ output @ its error

CBoW - Slide Context Window



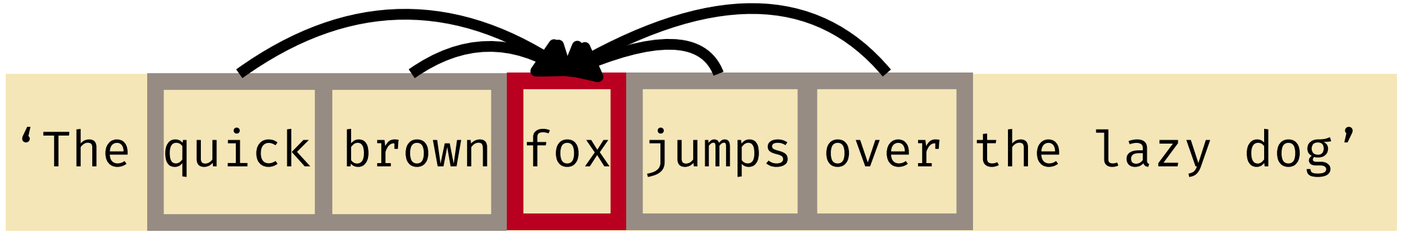
	E input					E’ output								
	E_0	E_1	E_2	E_3	E_4	E'_0	E'_1	E'_2	E'_3	E'_4	cores	y_hat	y_actual	error
*the	-0.13	0.45	0.23	0.10	-0.34	0.56	0.01	-0.32	0.37	-0.26	0.21	0.16	0.0	0.16
**quick	-0.46	-0.41	0.26	0.21	0.17	0.19	-0.19	-0.00	0.06	-0.32	-0.02	0.13	1.0	-0.87
*brown	-0.60	0.50	0.22	-0.18	-0.36	0.50	0.28	0.42	0.40	0.10	-0.09	0.12	0.0	0.12
*fox	-0.32	-0.20	0.02	-0.07	-0.21	0.44	-0.41	-0.32	-0.44	-0.17	-0.23	0.10	0.0	0.10
jumps	0.11	-0.36	-0.21	-0.13	-0.04	0.08	-0.23	0.30	-0.13	-0.22	0.09	0.14	0.0	0.14
over	0.29	-0.30	0.01	0.09	-0.45	0.07	-0.36	0.27	-0.42	0.49	-0.20	0.11	0.0	0.11
lazy	0.11	-0.33	-0.43	0.45	0.47	0.29	-0.30	-0.51	0.32	0.21	-0.34	0.09	0.0	0.09
dog	0.31	-0.20	-0.40	0.18	-0.06	0.25	0.27	-0.45	-0.13	-0.38	0.03	0.14	0.0	0.14

CBoW - Slide Context Window



	E input					E’ output								
	E_0	E_1	E_2	E_3	E_4	E'_0	E'_1	E'_2	E'_3	E'_4	scores	y_hat	y_actual	error
***the**	-0.12	0.43	0.24	0.11	-0.38	-0.53	-0.01	-0.33	0.37	-0.24	0.12	0.14	0.0	0.14
***quick**	-0.46	-0.41	0.26	0.21	0.17	0.03	-0.08	0.07	0.03	-0.45	0.07	0.13	0.0	0.13
****brown**	-0.59	0.48	0.23	-0.18	-0.40	0.52	0.26	0.41	0.40	0.12	-0.11	0.11	1.0	-0.89
***fox**	-0.31	-0.22	0.03	-0.06	-0.25	0.46	-0.42	-0.33	-0.44	-0.15	-0.05	0.12	0.0	0.12
***jumps**	0.11	-0.36	-0.21	-0.13	-0.04	-0.05	-0.25	0.29	-0.13	-0.19	0.09	0.14	0.0	0.14
over	0.29	-0.30	0.01	0.09	-0.45	0.09	-0.37	0.27	-0.42	0.51	-0.02	0.12	0.0	0.12
lazy	0.11	-0.33	-0.43	0.45	0.47	0.31	-0.31	-0.51	0.33	0.23	-0.08	0.12	0.0	0.12
dog	0.31	-0.20	-0.40	0.18	-0.06	0.28	0.25	-0.46	-0.13	-0.36	-0.08	0.12	0.0	0.12

CBoW - Slide Context Window



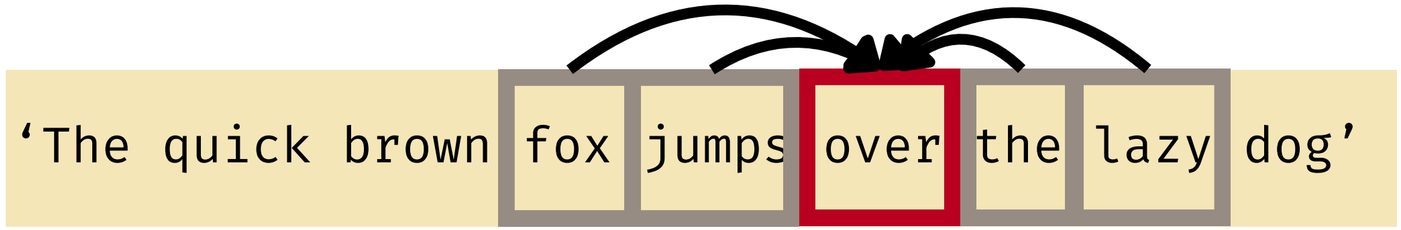
	E input					E’ output								
	E_0	E_1	E_2	E_3	E_4	E'_0	E'_1	E'_2	E'_3	E'_4	scores	y_hat	y_actual	error
the	-0.06	0.48	0.30	0.16	-0.35	-0.52	-0.00	-0.34	0.37	-0.23	0.08	0.14	0.0	0.14
***quick**	-0.41	-0.36	0.32	0.25	0.19	0.05	-0.07	0.06	0.03	-0.44	0.08	0.14	0.0	0.14
***brown**	-0.59	0.48	0.23	-0.18	-0.40	0.43	0.20	0.44	0.41	0.07	-0.04	0.12	0.0	0.12
****fox**	-0.25	-0.17	0.09	-0.01	-0.22	0.47	-0.42	-0.33	-0.44	-0.14	-0.03	0.12	1.0	-0.88
***jumps**	0.16	-0.31	-0.15	-0.08	-0.02	-0.04	-0.24	0.29	-0.13	-0.19	0.09	0.14	0.0	0.14
***over**	0.29	-0.30	0.01	0.09	-0.45	0.10	-0.37	0.26	-0.42	0.52	-0.04	0.12	0.0	0.12
lazy	0.11	-0.33	-0.43	0.45	0.47	0.32	-0.30	-0.52	0.33	0.23	-0.09	0.11	0.0	0.11
dog	0.31	-0.20	-0.40	0.18	-0.06	0.29	0.26	-0.46	-0.13	-0.35	-0.06	0.12	0.0	0.12

CBoW - Slide Context Window



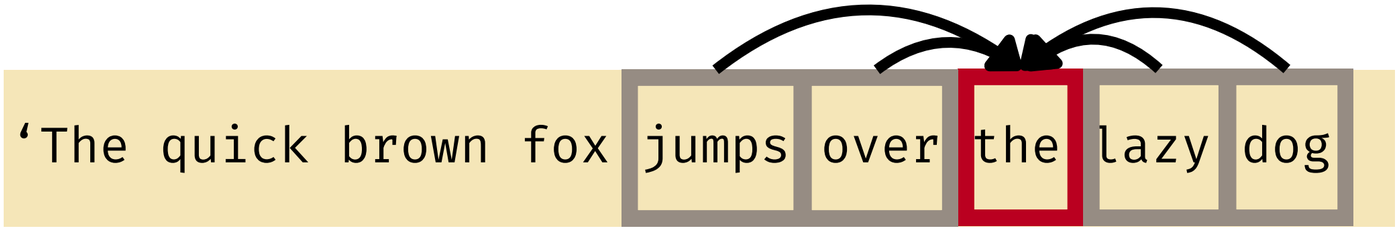
	E input					E’ output								
	E_0	E_1	E_2	E_3	E_4	E'_0	E'_1	E'_2	E'_3	E'_4	scores	y_hat	y_actual	error
*the	-0.06	0.48	0.30	0.16	-0.35	-0.51	0.00	-0.34	0.37	-0.22	0.09	0.14	0.0	0.14
quick	-0.37	-0.40	0.29	0.20	0.18	0.06	-0.06	0.05	0.03	-0.43	0.15	0.15	0.0	0.15
*brown	-0.54	0.45	0.20	-0.23	-0.41	0.44	0.21	0.44	0.41	0.08	-0.01	0.13	0.0	0.13
*fox	-0.25	-0.17	0.09	-0.01	-0.22	0.41	-0.47	-0.29	-0.43	-0.22	-0.06	0.12	0.0	0.12
**jumps	0.20	-0.35	-0.18	-0.14	-0.02	-0.03	-0.23	0.28	-0.13	-0.17	0.08	0.14	1.0	-0.86
*over	0.33	-0.34	-0.02	0.03	-0.46	0.11	-0.36	0.25	-0.42	0.53	-0.20	0.10	0.0	0.10
lazy	0.11	-0.33	-0.43	0.45	0.47	0.33	-0.30	-0.52	0.32	0.24	-0.24	0.10	0.0	0.10
dog	0.31	-0.20	-0.40	0.18	-0.06	0.30	0.27	-0.47	-0.13	-0.34	0.05	0.13	0.0	0.13

CBoW - Slide Context Window



	E input					E’ output								
	E_0	E_1	E_2	E_3	E_4	E'_0	E'_1	E'_2	E'_3	E'_4	scores	y_hat	y_actual	error
***the**	-0.08	0.47	0.34	0.14	-0.36	-0.50	-0.00	-0.35	0.37	-0.19	0.06	0.13	0.0	0.13
quick	-0.37	-0.40	0.29	0.20	0.18	0.07	-0.07	0.04	0.03	-0.40	0.02	0.13	0.0	0.13
brown	-0.56	0.43	0.24	-0.25	-0.42	0.45	0.20	0.43	0.41	0.10	0.00	0.12	0.0	0.12
***fox**	-0.27	-0.18	0.13	-0.03	-0.23	0.42	-0.48	-0.29	-0.43	-0.20	0.01	0.12	0.0	0.12
***jumps**	0.20	-0.35	-0.18	-0.14	-0.02	-0.09	-0.19	0.34	-0.14	-0.33	0.01	0.12	0.0	0.12
****over**	0.32	-0.35	0.02	0.02	-0.47	0.12	-0.36	0.25	-0.42	0.55	-0.04	0.12	1.0	-0.88
***lazy**	0.11	-0.33	-0.43	0.45	0.47	0.33	-0.30	-0.53	0.32	0.26	0.07	0.13	0.0	0.13
dog	0.31	-0.20	-0.40	0.18	-0.06	0.30	0.26	-0.48	-0.13	-0.32	-0.02	0.12	0.0	0.12

CBoW - Slide Context Window



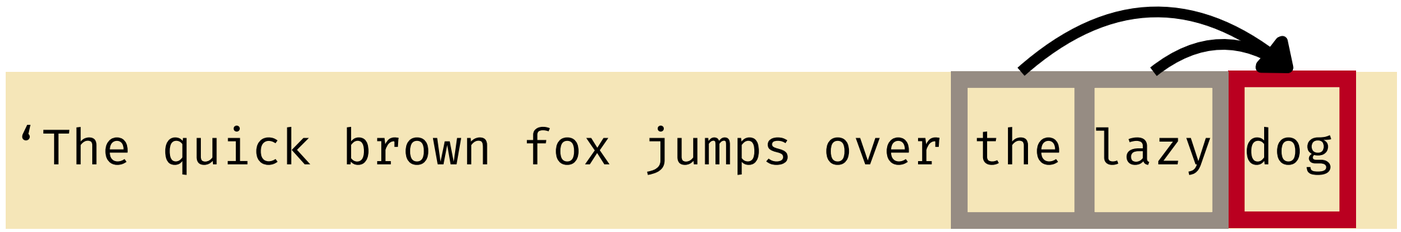
	E input					E’ output								
	E_0	E_1	E_2	E_3	E_4	E'_0	E'_1	E'_2	E'_3	E'_4	scores	y_hat	y_actual	error
**the	-0.09	0.44	0.39	0.08	-0.28	-0.50	0.00	-0.35	0.36	-0.19	-0.00	0.11	1.0	-0.89
quick	-0.37	-0.40	0.29	0.20	0.18	0.07	-0.07	0.05	0.02	-0.40	0.02	0.12	0.0	0.12
brown	-0.56	0.43	0.24	-0.25	-0.42	0.45	0.21	0.43	0.41	0.10	-0.02	0.11	0.0	0.11
fox	-0.28	-0.21	0.18	-0.09	-0.15	0.42	-0.47	-0.29	-0.44	-0.19	0.27	0.15	0.0	0.15
*jumps	0.20	-0.38	-0.14	-0.19	0.05	-0.09	-0.18	0.34	-0.14	-0.33	-0.06	0.11	0.0	0.11
*over	0.32	-0.35	0.02	0.02	-0.47	0.11	-0.41	0.23	-0.38	0.53	0.08	0.12	0.0	0.12
*lazy	0.11	-0.36	-0.39	0.40	0.55	0.33	-0.30	-0.53	0.32	0.26	0.33	0.16	0.0	0.16
*dog	0.31	-0.20	-0.40	0.18	-0.06	0.30	0.27	-0.48	-0.14	-0.31	0.07	0.12	0.0	0.12

CBoW - Slide Context Window



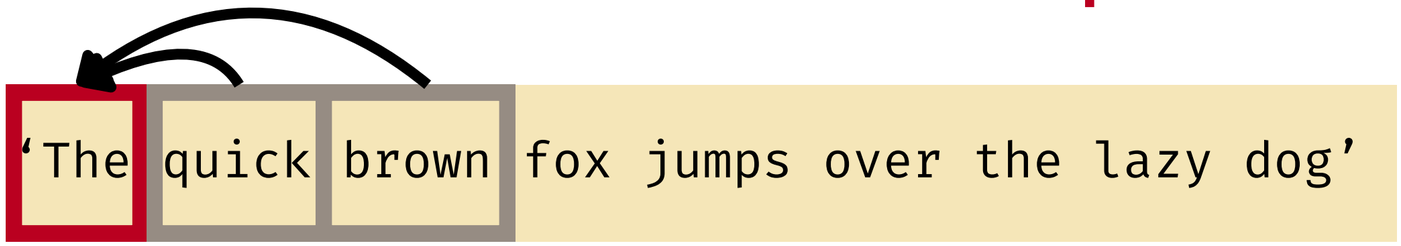
	E input					E’ output								
	E_0	E_1	E_2	E_3	E_4	E'_0	E'_1	E'_2	E'_3	E'_4	scores	y_hat	y_actual	error
*the	-0.09	0.44	0.39	0.08	-0.28	-0.39	-0.14	-0.45	0.41	-0.18	0.07	0.13	0.0	0.13
quick	-0.37	-0.40	0.29	0.20	0.18	0.05	-0.05	0.06	0.02	-0.40	0.12	0.14	0.0	0.14
brown	-0.56	0.43	0.24	-0.25	-0.42	0.43	0.23	0.44	0.40	0.10	0.06	0.13	0.0	0.13
fox	-0.28	-0.21	0.18	-0.09	-0.15	0.40	-0.45	-0.28	-0.45	-0.20	0.07	0.13	0.0	0.13
jumps	0.12	-0.36	-0.17	-0.14	0.04	-0.10	-0.16	0.35	-0.15	-0.33	0.06	0.13	0.0	0.13
*over	0.23	-0.34	-0.01	0.06	-0.48	0.10	-0.39	0.25	-0.38	0.53	-0.18	0.10	0.0	0.10
**lazy	0.03	-0.34	-0.42	0.44	0.53	0.31	-0.27	-0.51	0.31	0.26	0.02	0.12	1.0	-0.88
*dog	0.23	-0.18	-0.43	0.23	-0.08	0.29	0.29	-0.46	-0.15	-0.31	0.11	0.13	0.0	0.13

CBoW - Slide Context Window



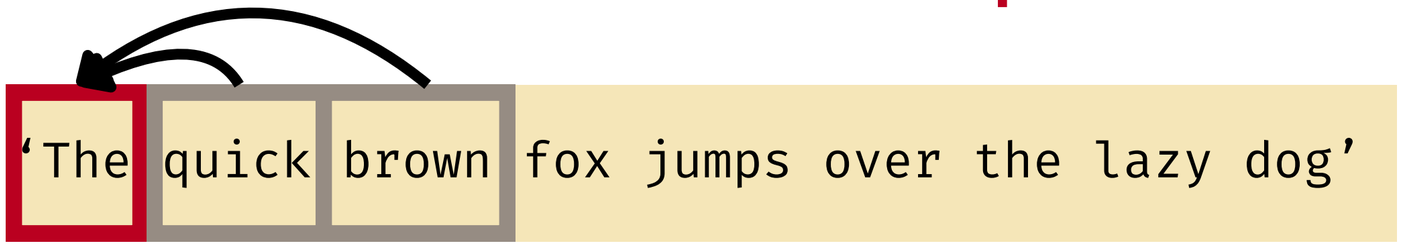
	E input					E’ output								
	E_0	E_1	E_2	E_3	E_4	E'_0	E'_1	E'_2	E'_3	E'_4	scores	y_hat	y_actual	error
***the**	-0.06	0.41	0.31	0.13	-0.23	-0.40	-0.14	-0.45	0.40	-0.17	0.12	0.14	0.0	0.14
quick	-0.37	-0.40	0.29	0.20	0.18	0.04	-0.05	0.06	0.01	-0.38	-0.06	0.12	0.0	0.12
brown	-0.56	0.43	0.24	-0.25	-0.42	0.43	0.23	0.45	0.39	0.12	0.11	0.14	0.0	0.14
fox	-0.28	-0.21	0.18	-0.09	-0.15	0.39	-0.44	-0.27	-0.46	-0.18	-0.16	0.11	0.0	0.11
jumps	0.12	-0.36	-0.17	-0.14	0.04	-0.11	-0.16	0.35	-0.16	-0.31	-0.11	0.11	0.0	0.11
over	0.26	-0.36	-0.08	0.11	-0.43	0.09	-0.39	0.25	-0.39	0.54	-0.06	0.12	0.0	0.12
***lazy**	0.03	-0.34	-0.42	0.44	0.53	0.37	-0.28	-0.52	0.36	0.14	0.14	0.14	0.0	0.14
****dog**	0.26	-0.21	-0.50	0.28	-0.02	0.28	0.29	-0.46	-0.15	-0.30	-0.06	0.12	1.0	-0.88

CBoW - After One Epoch



	E input					E’ output								
	E_0	E_1	E_2	E_3	E_4	E'_0	E'_1	E'_2	E'_3	E'_4	scores	y_hat	y_actual	error
**the	-0.02	0.51	0.22	0.09	-0.29	0.40	-0.14	-0.45	0.38	-0.18	0.08	0.14	1.0	-0.86
*quick	-0.37	-0.40	0.29	0.20	0.18	0.04	-0.05	0.06	-0.01	-0.39	0.04	0.14	0.0	0.14
*brown	-0.56	0.43	0.24	-0.25	-0.42	0.43	0.23	0.45	0.37	0.11	-0.10	0.12	0.0	0.12
fox	-0.28	-0.21	0.18	-0.09	-0.15	0.40	-0.45	-0.27	-0.47	-0.19	-0.23	0.11	0.0	0.11
jumps	0.12	-0.36	-0.17	-0.14	0.04	-0.11	-0.16	0.36	-0.17	-0.32	0.18	0.16	0.0	0.16
over	0.26	-0.36	-0.08	0.11	-0.43	0.09	-0.39	0.25	-0.41	0.53	-0.04	0.13	0.0	0.13
lazy	0.06	-0.24	-0.51	0.39	0.47	0.37	-0.29	-0.52	0.34	0.13	-0.34	0.10	0.0	0.10
dog	0.26	-0.21	-0.50	0.28	-0.02	0.28	0.30	-0.49	-0.03	-0.23	-0.22	0.11	0.0	0.11

CBoW - After 10 Epochs



	E input					E’ output								
	E_0	E_1	E_2	E_3	E_4	E'_0	E'_1	E'_2	E'_3	E'_4	cores	y_hat	y_actual	error
**the	1.42	1.95	0.48	0.17	0.06	-1.40	-1.54	-1.80	0.82	0.25	3.13	0.63	1.0	-0.37
*quick	-0.83	-1.39	-0.10	0.86	0.56	0.45	1.23	0.91	-0.59	-0.98	-0.10	0.02	0.0	0.02
*brown	-1.67	-0.08	-0.15	-0.51	-1.46	0.34	-0.31	1.44	0.31	0.72	-0.64	0.01	0.0	0.01
fox	-0.14	0.18	1.99	-0.70	0.02	0.54	-1.95	-0.37	-0.48	-0.59	2.33	0.28	0.0	0.28
jumps	0.01	-1.36	0.51	-0.53	1.05	0.08	0.24	0.89	-0.66	-1.56	0.20	0.03	0.0	0.03
over	0.55	-1.31	-0.67	0.13	-1.33	0.73	0.01	0.77	-0.76	1.52	-1.84	0.00	0.0	0.00
lazy	-0.42	0.22	-1.64	0.43	1.67	1.47	-0.23	-1.25	1.00	-0.90	-0.95	0.01	0.0	0.01
dog	0.06	-0.73	-1.63	1.29	-0.22	0.87	1.59	-1.19	0.38	1.02	-2.51	0.00	0.0	0.00

Next Week: Classification with ML!

